Aarón Aranda Torrijos



Python developer · Local-first software, lightweight and finished

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github.com/erlanders177 · linkedin.com/in/aaron-aranda-torrijos-94628b425 · pypi.org/project/bioforge

PROFESSIONAL SUMMARY

I'm 16 and I've been programming since 2024. In that time I've published a bioinformatics package on PyPI, a scientific calculator that works offline in the browser and on desktop, and a voice assistant that runs locally. All three share the same obsession: software that runs on modest hardware without sending data anywhere. What I do best is breaking a large problem into solvable pieces and not letting go until it works.

FEATURED PROJECTS

BioForge

Python, C, HTML

A bioinformatics engine built for modest, offline machines: it processes whole genomes without any data leaving the device.

- Uses 6.9× less memory than Biopython when loading FASTQ (115 MB vs 801 MB) thanks to a custom 5-bit encoding.

https://github.com/erlanders177/bioforge

Axioma

Python, Kotlin, JavaScript

A Spanish-language scientific calculator that works offline in the browser, on mobile and on desktop from a single codebase.

- One codebase for three platforms, running Python in the browser via Pyodide on WebAssembly and PyQt5 on desktop.

https://github.com/erlanders177/Axioma

Sakura

Python, JavaScript, CSS

A voice desktop assistant that automates computer tasks with a local model, reaching the cloud only when you ask it to.

- Two-tier architecture: local Ollama for actions like playing music, searching or sorting email; optional Groq for richer answers. The user decides how much leaves the machine.

https://github.com/erlanders177/sakura-asistente

EXPERIENCE & LEADERSHIP

Deputy lead · Club de ajedrez La Era de la Reina

09/2024 – Present

Volunteering · Madrid, Spain

- Overall champion of the 3rd La Era de la Reina school tournament (2025).
- Several seasons competing in the Madrid regional chess league.

Team technical lead · Certamen de robótica escolar

01/2025 – 06/2025

Academic

- Proposed the technical approach the team adopted as the most viable one, and led the execution, guiding my teammate.

TECHNICAL SKILLS

Languages: Python (intermediate), HTML y CSS (basic), C (basic)

Frameworks & libraries: PyQt5 (intermediate), NumPy (intermediate), SymPy (intermediate), FastAPI (basic), Electron (basic), Pyodide y WebAssembly (intermediate)

Areas of work: Software local y offline, Bioinformática, Voz y asistentes, LLM aplicados, Cálculo simbólico

Tools: Git y GitHub (intermediate), pytest (intermediate), GitHub Actions (intermediate), Linux (Ubuntu) (basic)

EDUCATION

Baccalaureate in Science and Technology

09/2026 – 06/2028

In progress. Expected graduation in 2028.

Secondary education certificate, with merit diploma

09/2022 – 06/2026

Standout grades: Technology (10), Research Project (9), Biology (8).

LANGUAGES

Spanish: Native · English: Intermediate · technical reading and written email

AWARDS

Overall champion, 3rd La Era de la Reina school tournament · Club de ajedrez La Era de la Reina · 06/2025